

Energy Efficient Data Transmission in WSN for Network Lifetime Enhancement

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Abstract: - Wireless Sensor Network (WSN) is made up of the multiple sensor nodes which can communicate with each other over a network wirelessly. The sensor node has significant use in the application of WSN. WSNs detect physical conditions of patients to sense various physical aspects and relay the critical information to the base station using the multi-hop data transmission technology. WSNs gather the information on the environment and physical world. In the Internet of Things (IoT) technology with numerous sensor nodes, WSNs are employed via the internet to collect the data. The energy supplied by battery is deemed as the important element of sensor nodes. Energy is what is needed one to complete a number of tasks to achieve efficient data delivery and transmission. The sensor nodes have battery lives with limited energy leading to the consideration of network life improvement. The network lifetime is one that is critical to WSN in ensuring the sustainable operation in remote places. A lot of machine learning techniques have been formulated in enhancing the lifetime of a network. However, the up-to-date technique did not improve the energy efficiency and the lifetime of the network. To overcome these problems, various clustering and optimization schemes are elaborated in regard to node localization schemes.

Keywords: *Wireless Sensor Network, Internet of Things, optimization, data delivery, data transmission, machine learning.*

I. INTRODUCTION

WSN is a separate solution that has number of sensor nodes to communicate wirelessly among themselves. WSN find application in various applications such as the home automation, forest control, health system and precision farming. Localization of sensor nodes is a vital component in WSN application. WSNs are used to determine the information and send to the Base Station (BS) by use of data transmission technology. The IoT technology employs WSNs in making data collection with large number of sensor nodes. The first one is the network lifetime enhancement, which is necessary in the context of WSN to enhance the sustainable work of sensor nodes in the distant points. Lots of researchers conducted their work as the need to optimize the energy consumption to increase the network lifetime. Data aggregation schemes scheme of classification and clustering has provided solution to the redundancy of data and tap energy. The necessary wireless sensor network process is represented in the figure 1.

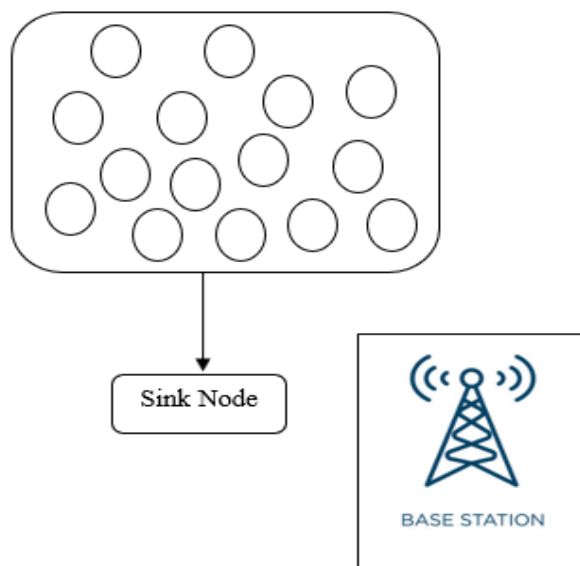


Fig.1: Wireless Sensor Network

The WSN topology is shown in figure 1. WSN has several sensor nodes, sink node and base station. The sensor nodes collect the data packets and send to the sink node and base station by accessing the internet. The sensor nodes also in WSN topology are utilized in the sensing region and relay the data to the distant site. The energy is significant in WSN through collection and transmission. The lifetime of the WSN is increased by reducing the consumption of energy.

The key input of the survey paper lies in dealing with main challenges of WSN issues. It tries to investigate ways of improving network life-span in data transmission that is essential in increasing the overall efficiency and life of the WSNs. Again, the article refers to different energy efficient methods of data aggregation but particularly focuses on those methods that control the time complexities to the minimum, thus maximizing the performance of the system. Lastly, it informs on the various energy-efficient data transmission modes which seek to minimize the use of energy which give an insight on the means of having a more sustainable and effective operations of the WSNs.

This paper is structured as follows; Section 1 gives the introduction to energy efficient data transmission in WSN. Section 2 presents the review of various methods for energy efficient data communication in WSN. Section 3 gives the dataset description. Section 4 explains the methodology for energy efficient data transmission with neat architecture diagram. Section 5 describes the performance metrics for energy efficient data transmission. Section 6 explains the result and discussion of diverse energy efficient data transmission methods. Section 7 concludes the review work.

II. LITERATURE SURVEY

In [1], the authors proposed DBN-RP as a means of distributing data in an efficient manner in a wireless sensor network connecting 5G. The selection of the cluster head was done under MRFO. Throughput was not enhanced when there was efficient data transmission. In [2], ELPSO-PSO-BPNN was developed to locate the sensor node. The developed algorithm ensured the localization and minimized the localization error. However, there was no saving on time taken in sensor node identification.

In [3], TIOCHR technique was introduced in order to optimize the energy usage. TIOCHR method was based on the selection of cluster heads basing on the reliability of data packets as well as the rate of delivery of data packets. The data collection was not, however, conducted effectively. In [4], there was the design of energy efficient network with intelligent clustering performed to allow efficient clustering to be carried out in a random fashion. A smart Silhouette Index (SI) clustering score was implemented to enhance the different living nodes. However, the optimization of the sensor node was not done by efficient clustering.

In [5], CSDAM was created to enable energy consumption reduction and network lifelong improvement. However, CSDAM did not conduct efficient data aggregation. In [6], an EDAS was proposed to carry out cluster-based dependable data transmission to extend the range of network lifetime. However, EDAS scheme did not enhance the throughput. The network parameter was measured by the use of EDAS scheme, which was used to enhance the performance of data transmission.

Bat Optimization Algorithm (BOA) was designed in [7] for sensor node localization in WSN in efficient manner. BOA algorithm reduced the error rate and time consumption. The FDAS was designed in [8] to gather the patient data and to minimize the energy consumption with higher network lifetime. The suitable parent node was chosen through fuzzy logic. However, the FDAS system failed to enhance the heterogeneous WSNs performance.

An energy-efficient data monitoring and aggregation method was employed in [9] to categorize the classified data into diverse classes depended on different features. The designed method performed K-means clustering and Extrema Point (EP) data aggregation. However, the classification accuracy was not improved by designed method. Broad transfer learning network was designed in [10] to perform efficient battery lifetime in vital way. AI-related methods minimized the computational power. However, the energy consumption was not examined by broad transfer learning network.

A SAEPNN was presented in [11] to address energy consumption involved in distance between unknown and known nodes. But, the mean square error involved in the prediction was not reduced. Transfer learning technique was developed in [12] to decrease the network lifetime in WSN. The root mean square error was not reduced significantly. Deep reinforcement learning was introduced in [13] to guarantee the quality of sensing coverage. A novel multistage exploration strategy was employed to increase the network depending on network state. However, the time consumption was not reduced by deep reinforcement learning concept.

Enhanced energy optimization model was introduced in [14] to guarantee maximum efficiency and network lifetime. The designed network was configured through minimum energy consumption and improved accuracy. Lagrangean heuristic strategy was employed in [15] to enhance the lifetime in WSN.

But the time consumption was not saved, through heuristic strategy that was designed. A better Harris hawks optimization based clustering and routing algorithm in the form of fuzzy algorithm was presented in [16] to enhance the lifetime of the network. There was the use of an encoding mechanism that chose the most ideal cluster head. However, the algorithm that was developed did not even assure reliability in wireless network. To prevent the imbalance problems, [17] developed an EECHIGWO algorithm. EECHIGWO algorithm enhanced the energy efficiency, throughput and the stability of the network of the WSNs.

In [18], a correlation model was proposed, to screen the sensor nodes along the correlation value by clustering. However, designed model did not decrease the computational cost. In [19], an original strategy was developed to combine the PSO with fuzzy architecture. The intended strategy involved the fuzzy logic to manage the goals with development strategies.

In [20], an IBORSDFNL Model was developed to be able to identify the various ways of gathering data in WSN. IBORSDFNL Model conducted the route path discovery and route path selection. In [21], a mapping method concept was developed having a linear transformation to improve the network lifetime performance. The technique developed combined the wide learning and transfer learning ideas.

In [22], an energy efficient method was developed that will reduce the energy burned by sensor node in the data transmission. The help of Euclidean distance and forward non-redundant data helps the redundancy to be reduced. In [23], MCDM which uses K-means algorithm to effectively aggregate data and forward them to the sink node was presented. The uncertainty in WSN has been dealt with by the use of Einstein Weighted Averaging Aggregation (EWA) operator.

The efficiency of data aggregation is presented in [24] where ML-EOA has been introduced. Cluster Head (CH) selection was done through ANN. In [25], an intelligent healthcare was developed where the sensors were to obtain the real-time health status. An encryption algorithm that utilized a genetic level ensured that data being transmitted in the wireless channel was preserved in order to do away with uncomfortable data transmission.

III. DATASET DESCRIPTION

The network lifetime improvement process is analyzed with the help of two datasets: network feature and health monitoring system dataset. The dataset of the network feature is obtained with the help of measurements of WSN, the data are obtained with 10 sensor nodes and a sink node. The data contains different network properties, which include signal strength, link quality, noise floor, transmission rates, reception rates, path length of the routing path, and ratios of characters received. It also has sensor values like temperature, humidity, and voltage measurements, and the data of a node is saved in a different CSV file. The data is divided into observations of 500 consecutive entries to each node where the performance of the network is in detail. The health monitoring system dataset used the second in the analysis consists of 4,286 instances and 20 features. The data is the documentation of many health parameters, such as temperature, heart rate, pulse, blood pressure, respiratory rate, oxygen saturation, and pH, which are monitored using sensors located on the body of a patient. On this information, the patients are classified into three health conditions which include severe, mild and moderate (chronic). The health monitoring data is applied in artificial intelligence and treating chronic diseases.

IV. METHODOLOGY

WSNs apply in various operations in tracking, monitoring, transportation and healthcare. WSNs are utilized in order to enhance the sensor node existence. Sensors have a significant component in battery energy.

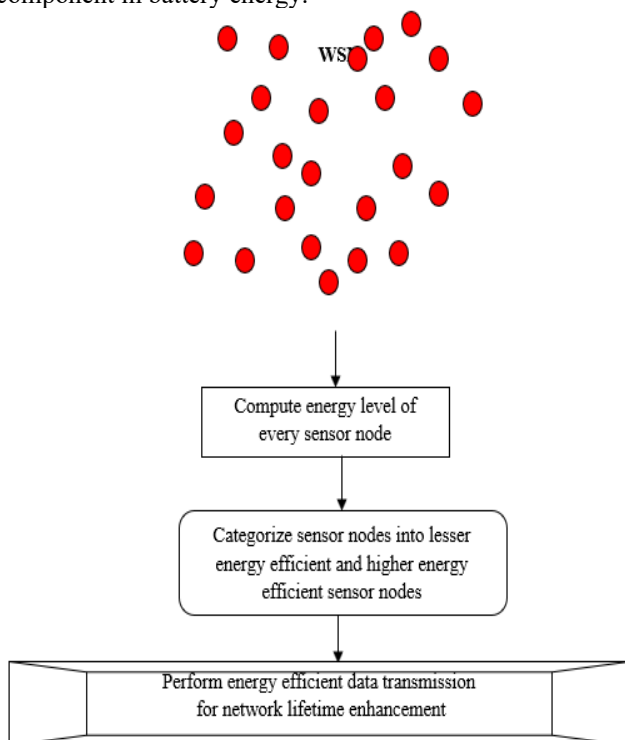


Fig.2: Energy Efficient Data Transmission

The reason why energy saving is necessary one is because there is lack of sufficient battery powering of the sensor nodes. One of these is energy which sensor nodes require to carry out different and diverse operations to facilitate efficient delivery and transmission of data, process and sense

data. When efficient data transmission occurs, it reduces the consumption of energy. Consequently, the distance on which Base Station (BS) and sensor node are located becomes smaller in order to save energy and prolong the lifetime of WSNs. The architecture diagram in WSN of energy efficient data transmission is shown in figure 2.

The way to transmit data in WSN is energy efficient as depicted in figure 2. The number of sensor nodes constitutes wireless sensor network. The energy level of each sensor nodes is established in order to do effective transmission of data. Subsequently, sensor nodes are classified as higher energy efficient sensor nodes and lesser energy efficient sensor node in WSN. The data transmission is done to ensure the enhancement of lifetime of the network with a high energy efficient sensor node.

A. Deep Learning-Based Routing Protocol

The Internet of Things (IoT) limited WSN is utilized to do the effective resource utilization and service delivery. IoT carried out effective network communication when the transmission of data between heterogeneous devices and energy-efficient WSN. The staging in WSN is power-saving and enhanced the network lifetime. WSN is also split into number of clusters during clustering process. To achieve effective data communication to the sink node by using selected CH, large consideration had to be taken by the cluster head (CH) selection. Besides, the node reachability is also augmented in cluster. Energy efficient DBN routing protocol is presented to make efficient transmission of data using selected route path. DBN based routing protocol is used to increase the packet delivery ratio (PDR). Wireless sensor network combines the nodes into as clusters with the help of the RL algorithm. The routing protocol of DBN made use of sensor node rewards of the particular cluster. The selection of cluster head in efficient communication of data is done based on MRFO algorithm.

B. Sensor Node Localization using Hybrid Particle Swarm Optimization

The localization problems are resolved with the help of WSN. Signal source identification in WSN is a key one in the emergency services. The range-based localization approach is also presented to solve the problem of sensor node localization. In maximum probability distribution function, the designed approach is employed. TDOA measurement model modeled as RSSI is proposed including Chan algorithm to determine the unknown node position. Hybrid particle swarm optimization localisation sensor node is an algorithm designed to work in WSN. In TDOA measurement model, ELPSO and PSO-BPNN was presented. TDOA model of measurement enhanced increased localization. The optimal distance method of localization minimized the error.

C. Trust Index Optimized Cluster Head Routing

WSN is utilized with the communication ability among the sensors. The possible path is chosen to ensure the trustworthiness and integrity. TIOCHR algorithm is applicable in augmenting the network lifetime and energy efficiency. The clustering stage included CH selection and recycling. The route selection and route identification is significant in routing stage. CH prevents the nodes that have the least degree of trust to limit the probability with network security. The safe channel of data transfer is selected in order to improve the network throughput. Trust metric based CH

and route selection is done to execute the energy conservation.

D. Cluster based Systematic Data Aggregation Model

A systematic data aggregation model in form of a cluster (called CSDAM) is presented to provide an efficient processing of real-time data. The cluster of active and sleep state nodes into the network is merged. The cluster- head (CH) is selected by the rank assigned to the sensors according to the amount of energy level (EL) and geographic location (GL) to the base station. CH is viewed as aggregator. The aggregation is done in three levels. The level 3 processing of data is done using the level 1 and level 2 data. Aggregation is done when aggregator falls below threshold.

E. Bat Optimization Algorithm based Sensor Node Localization in WSN

WSN consisted of the sensor nodes in huge amount of information in an attempt to gather information regarding adjacent sensor node. The WSN sensor node localization is significant in a military application. The localization problem is to determine the coordinates target node coordinates of the anchor nodes. BOA is presented to determine the sensor nodes in the most reasonable manner towards solving the local optimum. BOA features are varied in BOA variants 1 and 2 that feature global search strategy and local search strategy. BOA variant 1 and 2 minimised the error, restricted the count of nodes and the computing time. Variants 1 and 2 of the BOA increased the efficiency of localization of the various target and anchor nodes.

F. Fuzzy-Based Efficient Healthcare Data Collection

An Internet of Things (IoT) represents a high level of technology, and there are numerous devices to gather, convey, and receive information. Gathering of the data in health sector is necessary due to increased efficiency. In cases where the sensors produce big amount of data, it is necessary to collect the data so that it can be effectively transmitted. The redundant information was gathered with the help of sensors and transmitted similar values repeatedly. The data transmission involved negative impact on heterogeneous network performance outcome such as energy consumption increase and control overhead. The transmission slots will be significant in transmitting the data packets. The FDAS is proposed to aggregate the data packets as well as reduce the regular data size to enhance network performance. FDAS minimized the controversy of the energy use in the health sector due to redundant information. FDAS minimized the power use through a better aggregation. FDAS minimised the control overhead using minimum transmission slots.

G. Deep Reinforcement Learning Approach

MBW deployed energy replenishment technology of the WSN. Mobile Charger (MC) is applied to charge the nodes with mobile charging scheduling outcome. Sensing coverage is a significant network attribute that draws much attention to mobile charging scheduling. The charging requirements were dynamic at different time period. Online mobile charging scheduling took critical position relative to certain network model that was to be utilized in actual practice. An OMCS-QSC algorithm of WRSN is proposed through a model-free deep reinforcement algorithm. The online charging schedules are determined with the help of MC that is proposed as the agent by using multistage exploration policy. The use of OMCS-QSC is in increasing network QSC in line with the

real-time network condition. There is a reward function used to calculate MC charging action based on sensor node real time sensor coverage. Good convergence of large-scale WRSNs can be attained by MEDQN.

V. PERFORMANCE METRICS

The result analysis of existing energy efficient data transmission methods for network lifetime enhancement is carried out using four dissimilar parameters, namely: Energy consumption, Network lifetime, Scheduling time and Data aggregation accuracy.

A. Results of Energy Consumption

The amount of energy used is determined using the number of nodes that is taken into account during the data aggregation process and the amount of energy that is used in designing a sensor node to carry out the entire process. It is calculated in units of joules (Joule). It is computed as,

$$EC = N_i *$$

Energy(one sensor node for data aggregation) (1)

From equation (1), 'EC' denote the energy consumption. ' N_i ' symbolizes the total number of sensor nodes.

B. Results of Network Lifetime

Network Lifetime is a ratio of the number of sensor nodes addressed in order to provide efficient data aggregation in the network to the total sensor nodes taken into account in the simulation. Network lifetime is determined percentage (%). It is formulated as,

$$NL = \frac{N_{DA}}{N_i} * 100 \quad (2)$$

From equation (2), 'NL' symbolizes the network lifetime. ' N_{DA} ' represents the number of sensor nodes addressed. ' N_i ' symbolizes the total number of sensor nodes.

C. Results of Scheduling Time

The time during which the sensor node in WSN takes in data aggregation to do energy efficient data aggregation is called scheduling time and is defined as the product of number of sensor nodes and time in sensor node. The time scheduling is done based on milliseconds (ms). It is calculated as,

$$ST = N_i * \text{Time}(\text{Scheduling one sensor node}) \quad (3)$$

From equation (3), 'ST' represent the scheduling time. ' N_i ' symbolizes the total number of sensor nodes.

D. Results of Data Aggregation Accuracy

Accuracy of data aggregation is calculated as proportion of the correctly aggregated data packets to the overall number of data packets that have been sent. It is also calculated in percentage (percentage). It is formulated as,

$$DAA = \frac{\text{Number of data packets correctly aggregated}}{\text{Total number of data packets sent}} * 100 \quad (4)$$

From (4), 'DAA' represent the data aggregation accuracy. When the data aggregation accuracy is high, the method is said to be more efficient.

VI. RESULTS AND DISCUSSION

The outcome achieved is founded on energy efficient transmission of data in WSN under various methods. It is done through the analysis of performance with the assistance of energy efficient data transmission in WSN. Table and graphs assist in discussing the performance of the existing energy efficient data transmission in WSN.

Table.I. Tabulation on Data Aggregation Accuracy, Scheduling Time, Energy Consumption and Network Lifetime for Network Feature Dataset

Methods/Techniques	Energy Consumption (J)	Network Lifetime (%)	Data Aggregation Accuracy (%)	Scheduling Time (ms)
DBN based routing protocol [1]	0.4	95	91	17
ELPSO-PSO-BPNN [2]	0.8	92	95	19
TIOCHR algorithm [3]	0.35	99	98	13
CSDAM [5]	45	81	85	20
EDAS [6]	97	70	50	102
BOA [7]	68	52	75	55
FDAS [8]	40	85	81	20
SAE-PNN [11]	0.5	98	96	17
Model-free deep reinforcement learning [13]	18	82	80	29

Table.II. Tabulation on Data Aggregation Accuracy, Scheduling Time, Energy Consumption and Network Lifetime for Health Monitoring System Dataset

Methods/Techniques	Energy Consumption (J)	Network Lifetime (%)	Data Aggregation Accuracy (%)	Scheduling Time (ms)
DBN based routing protocol [1]	0.43	96	89	19
ELPSO-PSO-BPNN [2]	0.91	91	93	21
TIOCHR algorithm [3]	0.4	98	95	15
CSDAM [5]	48.2	78	82	23
EDAS [6]	99.1	68	48	104
BOA [7]	70.8	49	72	58
FDAS [8]	43.8	83	79	22
SAE-PNN [11]	0.8	95	94	18
Deep reinforcement learning [13]	21.8	80	78	31

Table I and Table II describes the performance results for nine dissimilar methods with four performance metrics and two datasets correspondingly. From above table, it is clear that TIOCHR algorithm attained improved results than other existing energy efficient data aggregation techniques methods for three datasets. For Network Feature Dataset, TIOCHR algorithm attained 0.35 Joules of energy consumption, 98% of network lifetime, 99% of data aggregation accuracy and 13ms of scheduling time. For Health Monitoring System Dataset, TIOCHR algorithm attained 0.4 Joules of energy consumption, 96% of network lifetime, 95% of data aggregation accuracy and 15ms of scheduling time. Figure 3, and figure 4 illustrate the performance analysis of Network Feature Dataset and Health Monitoring System Dataset.

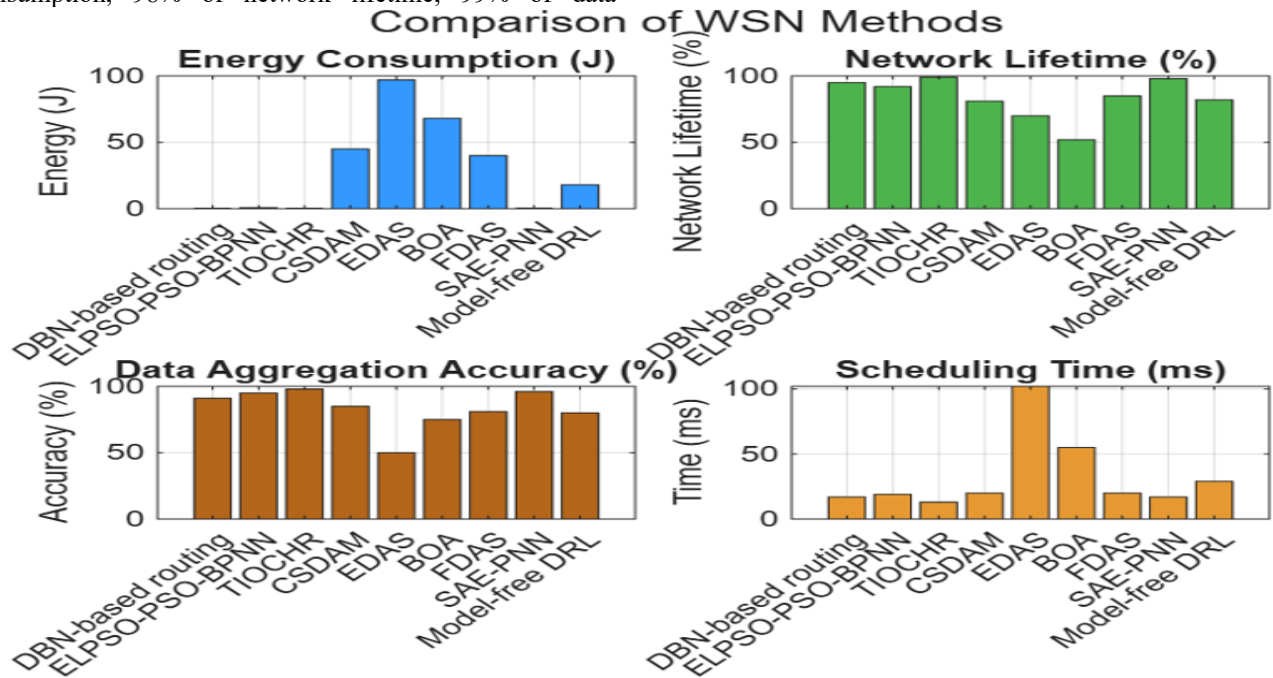


Fig.3: Performance Analysis on Energy Consumption, Network Lifetime, Data Aggregation Accuracy and Scheduling Time for Network Feature Dataset

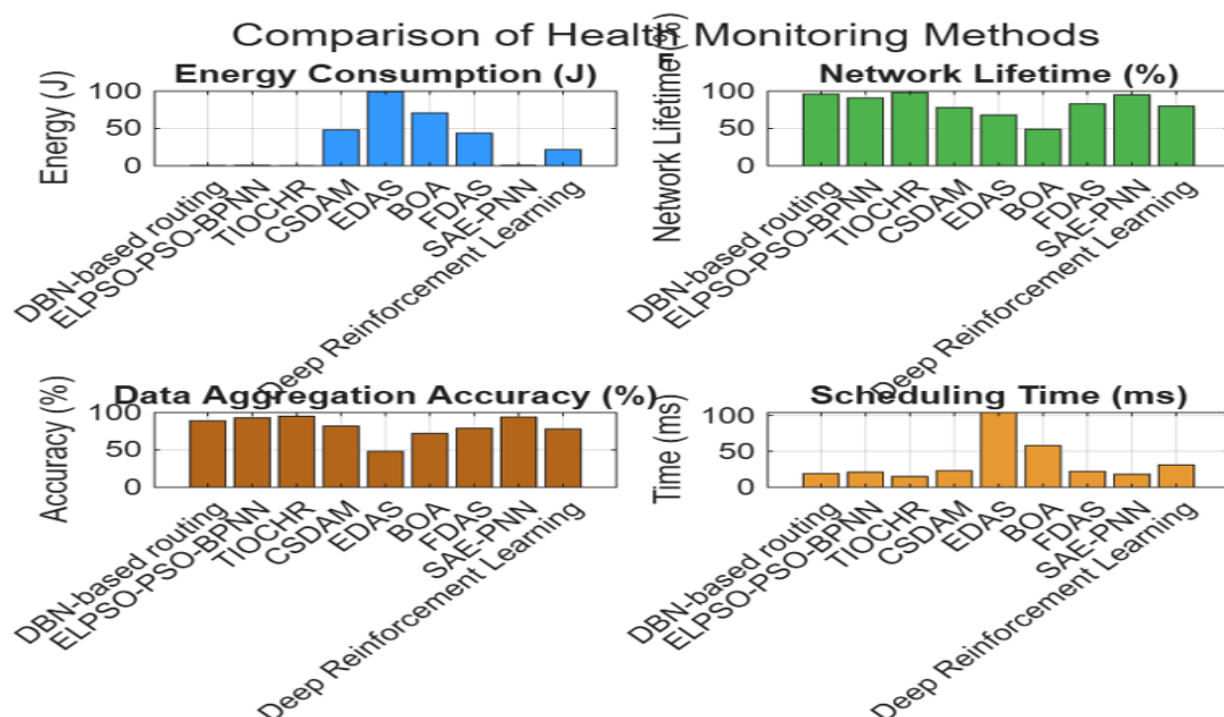


Fig.4: Performance Analysis on Energy Consumption, Network Lifetime, Data Aggregation Accuracy and Scheduling Time for Health Monitoring System Dataset

Figure 3 and figure 4 illustrates the performance metric comparison of nine different existing methods for two different datasets correspondingly. From graphical analysis, it observed that performance of TIOCHR algorithm is higher than any other techniques for three datasets. This is because of using cluster head selection with energy backup, packet delivery rate and packet reliability. TIOCHR algorithm increased energy efficiency and network lifetime. The route identification and route selection are employed in routing phase. CH and route selection is carried out for performing the energy conservation with minimum time complexity. For Network Feature Dataset, TIOCHR algorithm reduced energy consumption by 74%, increased network lifetime by 26%, increased data aggregation accuracy by 25% and minimized scheduling time by 46% when compared to other existing techniques. For Health Monitoring System Dataset, TIOCHR algorithm reduced energy consumption by 76%, increased network lifetime by 28%, increased data aggregation accuracy by 25% and reduced scheduling time by 43% when compared to other state-of-the-art methods.

VII. CONCLUSION

In the survey, energy efficient data transmission in WSN is carried out with network feature and healthcare monitoring datasets using different techniques. Different performance metrics are used to analyze the energy efficient data transmission with number of data packets and sensor nodes. When the result is evaluated with another study, energy efficient data transmission with deep learning methods has attained better results. In this work, TIOCHR algorithm attained higher data aggregation accuracy for three different datasets. Cluster head selection is carried out to achieve the improved accuracy result than other conventional techniques. These results strongly suggest deep learning concepts can be

implemented in future for energy efficient data transmission in WSN instead of the other conventional methods.

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